

Lab Experiments

Set 1: Monthly Sales Data

R PROGRAM:

```
# Monthly Sales Dataset
month <- c("Jan","Feb","Mar","Apr","May")
sales <- c(15000,18000,22000,20000,23000)
advertising <- c(2000,2500,3000,2800,3500)
```

```
# Display 3 graphs together
par(mfrow=c(1,3))
```

```
# 1. Line Chart
```

```
plot(sales,
     type="o",
     col="blue",
     xaxt="n",
     main="Sales Trend",
     xlab="Month",
     ylab="Sales")
axis(1,at=1:5,labels=month)
```

```
# 2. Bar Chart
```

```
barplot(sales,
        names.arg=month,
        col="green",
        main="Sales Comparison",
        xlab="Month",
        ylab="Sales")
```

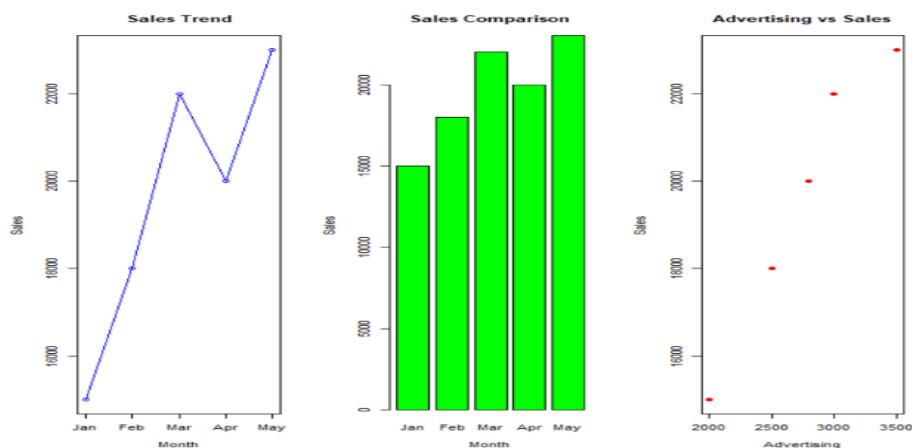
```
# 3. Scatter Plot
```

```
plot(advertising,sales,
     pch=19,
     col="red",
     main="Advertising vs Sales",
     xlab="Advertising",
     ylab="Sales")
```

```
# Reset Layout
```

```
par(mfrow=c(1,1))
```

OUTPUT:



Set 2: Customer Feedback Analysis

R PROGRAM:

```
# Customer Feedback Dataset
age <- c(25,30,35,28,40,32,27,29,38,45)
satisfaction <- c(4,5,3,4,5,4,3,5,4,5)

# Display 4 graphs together
par(mfrow=c(2,2))

# 1. Histogram
hist(age,
     col="skyblue",
     main="Age Distribution",
     xlab="Age",
     ylab="Frequency")

# 2. Pie Chart
pie(table(satisfaction),
    col=rainbow(length(table(satisfaction))),
    main="Customer Satisfaction Ratings")

# 3. Bar Chart
barplot(table(satisfaction),
        col="lightgreen",
        main="Satisfaction Frequency",
        xlab="Ratings",
        ylab="Count")

# 4. Scatter Plot
plot(age,satisfaction,
     pch=19,
     col="red",
     main="Age vs Satisfaction",
     xlab="Age",
     ylab="Satisfaction Score")

# Reset Layout
par(mfrow=c(1,1))
```

OUTPUT:



R PROGRAM:

```
# Employee Performance Dataset
employee_id <- c(1,2,3,4,5)
department <- c("Sales","HR","Marketing","Sales","HR")
years_service <- c(5,3,7,4,2)
performance <- c(85,92,78,90,76)

# Display 3 graphs together
par(mfrow=c(1,3))

# 1. Line Chart
plot(years_service, performance,
     type="o",
     col="blue",
     pch=16,
     main="Employee Performance Trend",
     xlab="Years of Service",
     ylab="Performance Score")

legend("bottomright",
     legend="Performance",
     col="blue",
     lty=1,
     pch=16)

# 2. Bar Chart
dept_count <- table(department)

barplot(dept_count,
     col="green",
     main="Employees by Department",
     xlab="Department",
     ylab="Number of Employees")

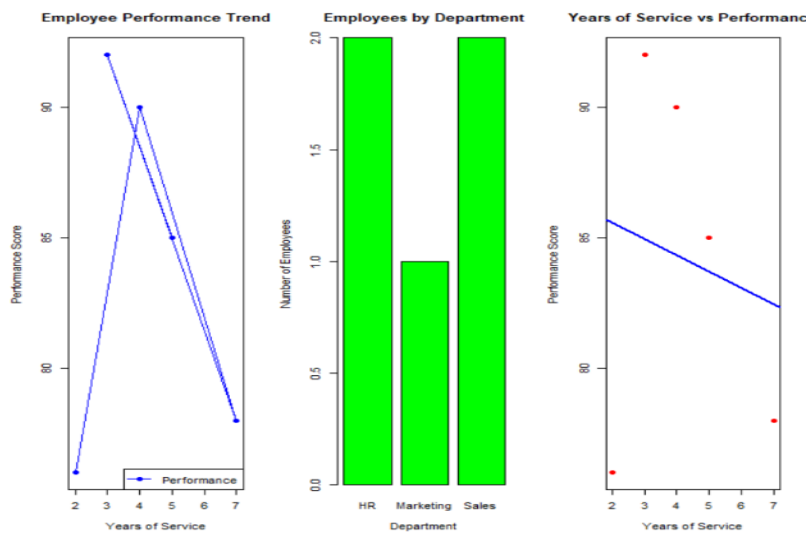
# 3. Scatter Plot
plot(years_service, performance,
     pch=19,
     col="red",
     main="Years of Service vs Performance",
     xlab="Years of Service",
     ylab="Performance Score")

# Best Fit Line
abline(lm(performance ~ years_service),
     col="blue",
     lwd=2)

# Reset Layout
par(mfrow=c(1,1))

# Scatter Plot Interpretation
cat("Interpretation:\n")
cat("The scatter plot shows the relationship between years of service and performance score.\n")
cat("Employees with more years of service generally tend to have better performance scores.\n")
```

OUTPUT:



Set 4: Product Inventory Management

R PROGRAM:

```
# Product Inventory Dataset
product <- c("Product A","Product B","Product C","Product D","Product E")
quantity <- c(250,175,300,200,220)
price <- c(500,400,600,450,550)

# Display 3 graphs together
par(mfrow=c(1,3))

# 1. Bar Chart
barplot(quantity,
  names.arg=product,
  col="skyblue",
  main="Product Quantity",
  xlab="Products",
  ylab="Quantity Available")

# 2. Stacked Bar Chart
category <- c("Category 1","Category 1","Category 2","Category 2","Category 3")
inventory <- table(category, product)

barplot(inventory,
  col=rainbow(nrow(inventory)),
  main="Inventory by Category",
  xlab="Products",
  ylab="Quantity",
  legend=rownames(inventory))

# 3. Scatter Plot
plot(price, quantity,
  pch=19,
  col="red",
  main="Price vs Quantity",
  xlab="Product Price",
```

```

ylab="Quantity Available")

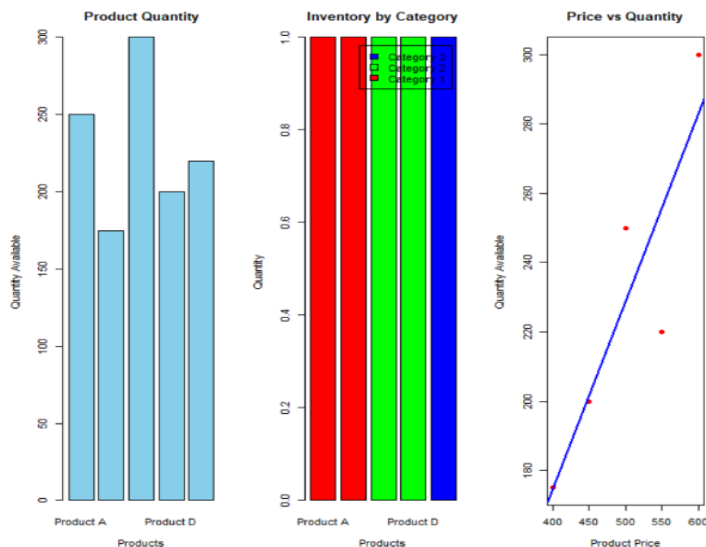
# Best Fit Line
abline(lm(quantity ~ price),
      col="blue",
      lwd=2)

# Reset Layout
par(mfrow=c(1,1))

# Scatter Plot Interpretation
cat("Interpretation:\n")
cat("The scatter plot shows the relationship between product price and quantity available.\n")
cat("Products with higher prices may have different inventory levels compared to lower-priced\n")
cat("products.\n")

```

OUTPUT:



Set 5: Website Analytics

R PROGRAM:

```

# Website Traffic Dataset
date <- c("2023-01-01", "2023-01-02", "2023-01-03", "2023-01-04", "2023-01-05")
page_views <- c(1500, 1600, 1400, 1650, 1800)
ctr <- c(2.3, 2.7, 2.0, 2.4, 2.6)

# User Interaction Data (Sample)
likes <- c(300, 350, 280, 400, 450)
shares <- c(120, 150, 100, 130, 170)
comments <- c(60, 70, 50, 80, 90)

# Display 3 graphs together
par(mfrow=c(1,3))

# 1. Line Chart

```

```

plot(page_views,
     type="o",
     col="blue",
     xaxt="n",
     main="Daily Page Views",
     xlab="Date",
     ylab="Page Views")
axis(1, at=1:5, labels=date)

```

2. Bar Chart

```

barplot(ctr,
       names.arg=date,
       col="green",
       main="Click-through Rate",
       xlab="Date",
       ylab="CTR (%)")

```

3. Stacked Area Chart

```

interaction <- rbind(likes, shares, comments)

```

```

x <- 1:5

```

```

plot(x, likes,
     type="n",
     ylim=c(0, max(colSums(interaction))),
     xaxt="n",
     xlab="Date",
     ylab="Interactions",
     main="User Interactions")

```

```

axis(1, at=1:5, labels=date)

```

```

polygon(c(x, rev(x)),
       c(rep(0,5), rev(likes)),
       col="lightblue",
       border=NA)

```

```

polygon(c(x, rev(x)),
       c(likes, rev(likes+shares)),
       col="lightgreen",
       border=NA)

```

```

polygon(c(x, rev(x)),
       c(likes+shares, rev(likes+shares+comments)),
       col="pink",
       border=NA)

```

```

legend("topleft",
     legend=c("Likes", "Shares", "Comments"),
     fill=c("lightblue", "lightgreen", "pink"))

```

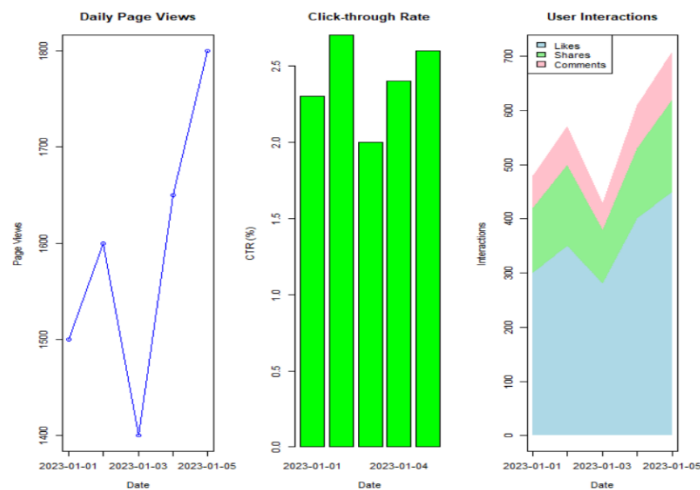
Reset Layout

```

par(mfrow=c(1,1))

```

OUTPUT:



```
# Monthly Product Sales Dataset
```

```
product <- c("Product A","Product B","Product C")
```

```
jan <- c(2000,1500,1200)
```

```
feb <- c(2200,1800,1400)
```

```
mar <- c(2400,1600,1100)
```

```
# Display 3 outputs together
```

```
par(mfrow=c(1,3))
```

```
# 1. Grouped Bar Chart
```

```
sales <- rbind(jan, feb, mar)
```

```
barplot(sales,  
  beside=TRUE,  
  col=c("red","green","blue"),  
  names.arg=product,  
  main="Quarter 1 Product Sales",  
  xlab="Products",  
  ylab="Sales",  
  legend.text=c("January","February","March"))
```

```
# 2. Stacked Area Chart
```

```
month <- c(1,2,3)
```

```
plot(month, jan,  
  type="n",  
  xaxt="n",  
  ylim=c(0, max(jan+feb+mar)),  
  xlab="Month",  
  ylab="Sales",  
  main="Overall Sales Trend")
```

```
axis(1,  
  at=1:3,  
  labels=c("January","February","March"))
```

```
polygon(c(month, rev(month)),  
  c(rep(0,3), rev(jan)),  
  col="lightblue",  
  border=NA)
```

```

polygon(c(month, rev(month)),
        c(jan, rev(jan+feb)),
        col="lightgreen",
        border=NA)

polygon(c(month, rev(month)),
        c(jan+feb, rev(jan+feb+mar)),
        col="pink",
        border=NA)

legend("topleft",
       legend=product,
       fill=c("lightblue","lightgreen","pink"))

```

```
# 3. Table
```

```

plot.new()
title("Monthly Sales Table")

```

```

table_data <- data.frame(
  Product=product,
  January=jan,
  February=feb,
  March=mar
)

```

```

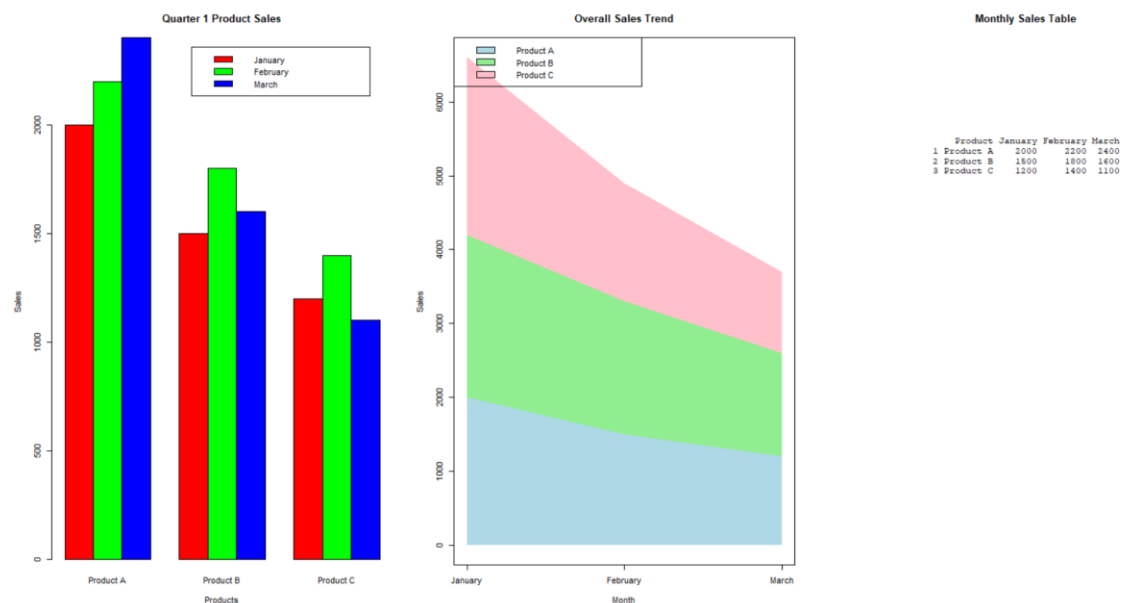
text(0.5,0.8,
     paste(capture.output(print(table_data)),
           collapse="\n"),
     family="mono")

```

```
# Reset Layout
```

```
par(mfrow=c(1,1))
```

OUTPUT:



Set 7: Customer Demographics Analysis

R PROGRAM:

```
# Customer Demographics Dataset
customer_id <- c(1,2,3)
age <- c(28,35,42)
gender <- c("Female","Male","Female")
income <- c(50000,60000,75000)

# Display 3 graphs together
par(mfrow=c(1,3))

# 1. Bar Chart
barplot(age,
        names.arg=customer_id,
        col="skyblue",
        main="Customer Age Distribution",
        xlab="Customer ID",
        ylab="Age")

# 2. Pie Chart
pie(table(gender),
    col=c("pink","lightblue"),
    main="Customer Distribution by Gender")

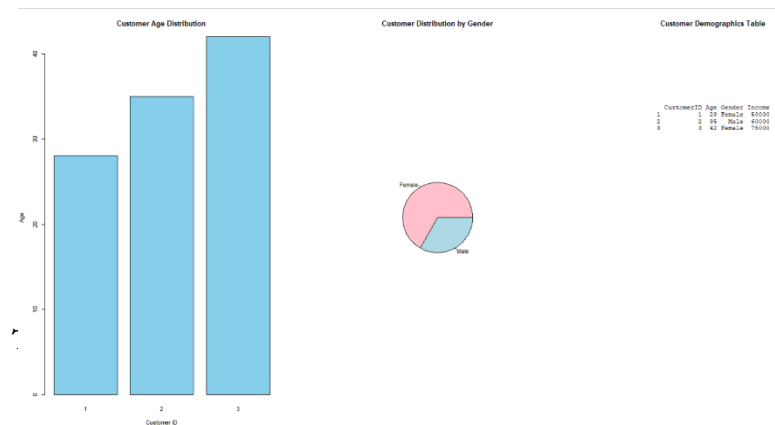
# 3. Table
plot.new()
title("Customer Demographics Table")

table_data <- data.frame(
  CustomerID=customer_id,
  Age=age,
  Gender=gender,
  Income=income
)

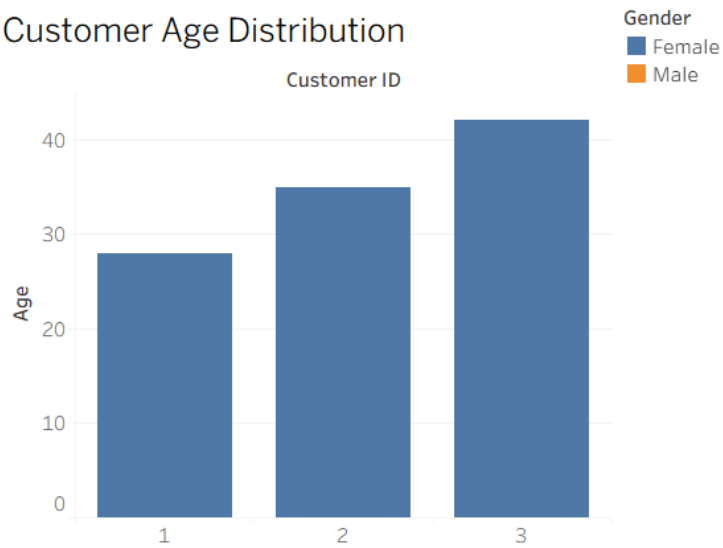
text(0.5,0.8,
     paste(capture.output(print(table_data)),
           collapse="\n"),
     family="mono")

# Reset Layout
par(mfrow=c(1,1))
```

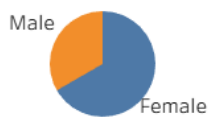
OUTPUT:



Customer Age Distribution



Customer Distribution by Gender



Set 8: Employee Performance Analysis

R PROGRAM

```
# Employee Performance Dataset
employee_id <- c(1,2,3)
department <- c("Sales","HR","Marketing")
years_service <- c(5,3,7)
performance <- c(85,92,78)
```

```
# Display 3 outputs together
par(mfrow=c(1,3))
```

```
# 1. Line Chart
plot(years_service, performance,
     type="o",
     pch=19,
     col="blue",
     main="Employee Performance Trend",
     xlab="Years of Service",
     ylab="Performance Score")
```

```
# 2. Bar Chart
dept_count <- table(department)
```

```
barplot(dept_count,
       col="lightgreen",
```

```

    main="Employees by Department",
    xlab="Department",
    ylab="Number of Employees")

# 3. Table
plot.new()
title("Employee Performance Table")

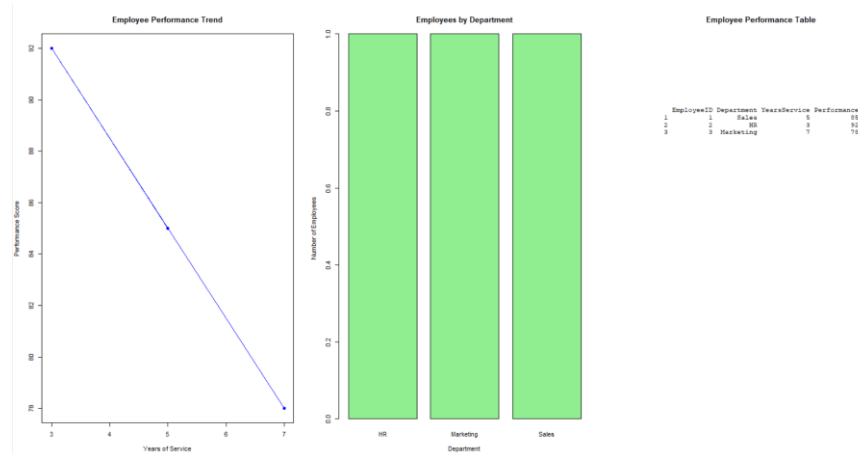
table_data <- data.frame(
  EmployeeID=employee_id,
  Department=department,
  YearsService=years_service,
  Performance=performance
)

text(0.5,0.8,
     paste(capture.output(print(table_data)),
           collapse="\n"),
     family="mono")

# Reset Layout
par(mfrow=c(1,1))

```

OUTPUT



Set 9: Online Learning Activity

R PROGRAM

```

student_id <- c("L01","L02","L03","L04","L05","L06")
gender <- c("Male","Female","Male","Female","Male","Female")
age <- c(20,22,19,21,23,20)
course <- c("R","R","SQL","R","R","SQL")
study_time <- c(3.5,4.2,2.0,5.0,2.5,4.0)
videos_watched <- c(12,15,8,18,9,14)

```

```
quiz_score <- c(78,85,65,92,70,88)
login_date <- as.Date(c("2025-01-05","2025-01-05",
                        "2025-02-08","2025-02-08",
                        "2025-03-12","2025-03-12"))
```

```
# Display 4 graphs together
par(mfrow=c(2,2))
```

```
# 1(a). Histogram
hist(quiz_score,
     col="skyblue",
     main="Histogram of Quiz Scores",
     xlab="Quiz Score",
     ylab="Frequency")
```

```
# 1(b). Boxplot by Course
boxplot(quiz_score ~ course,
       col=c("lightgreen","pink"),
       main="Quiz Score by Course",
       xlab="Course",
       ylab="Quiz Score")
```

```
# 2. Bubble Scatter Plot
symbols(study_time, quiz_score,
       circles=videos_watched/5,
       inches=0.3,
       bg=rgb(1,0,0,0.5),
       fg="black",
       xlab="Study Time (hrs)",
       ylab="Quiz Score",
       main="Study Time vs Quiz Score")
```

```
# 3. Monthly Average Quiz Score
month <- format(login_date,"%b")
avg_score <- tapply(quiz_score, month, mean)
```

```
plot(avg_score,
     type="o",
     col="blue",
     xaxt="n",
     xlab="Month",
     ylab="Average Quiz Score",
     main="Monthly Average Quiz Score")
```

```
axis(1,
     at=1:length(avg_score),
     labels=names(avg_score))
```

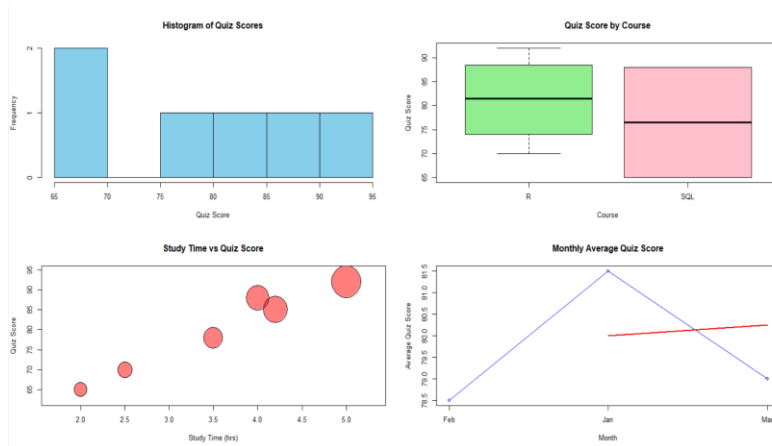
```
lines(stats::filter(avg_score, rep(1/2,2), sides=1),
     col="red",
     lwd=2)
```

```
# Reset Layout
par(mfrow=c(1,1))
```

```
# Interpretation
```

```
cat("Interpretation:\n")
cat("Histogram and boxplot summarize quiz performance.\n")
cat("Bubble plot shows that higher study time and more videos watched generally improve quiz scores.\n")
cat("The monthly line chart displays the average quiz score trend over time.\n")
```

OUTPUT



R PROGRAM:

```
# Survey Responses Dataset
survey_id <- c(1,2,3)
question1 <- c("A","B","C")
question2 <- c("B","A","A")
question3 <- c("C","D","B")

# Display 3 outputs together
par(mfrow=c(1,3))

# 1. Grouped Bar Chart (Question 1 Responses)
q1 <- table(question1)

barplot(q1,
        beside=TRUE,
        col="skyblue",
        main="Question 1 Responses",
        xlab="Answer",
        ylab="Frequency")

# 2. Stacked Bar Chart (All Questions)
response_data <- rbind(
  table(question1),
  table(factor(question2, levels=c("A","B","C","D"))),
  table(factor(question3, levels=c("A","B","C","D")))
)

barplot(response_data,
        beside=FALSE,
        col=rainbow(3),
        main="Survey Response Distribution",
        xlab="Answers",
```

```

      ylab="Frequency",
      legend.text=c("Question 1","Question 2","Question 3"))

# 3. Table
plot.new()
title("Survey Response Table")

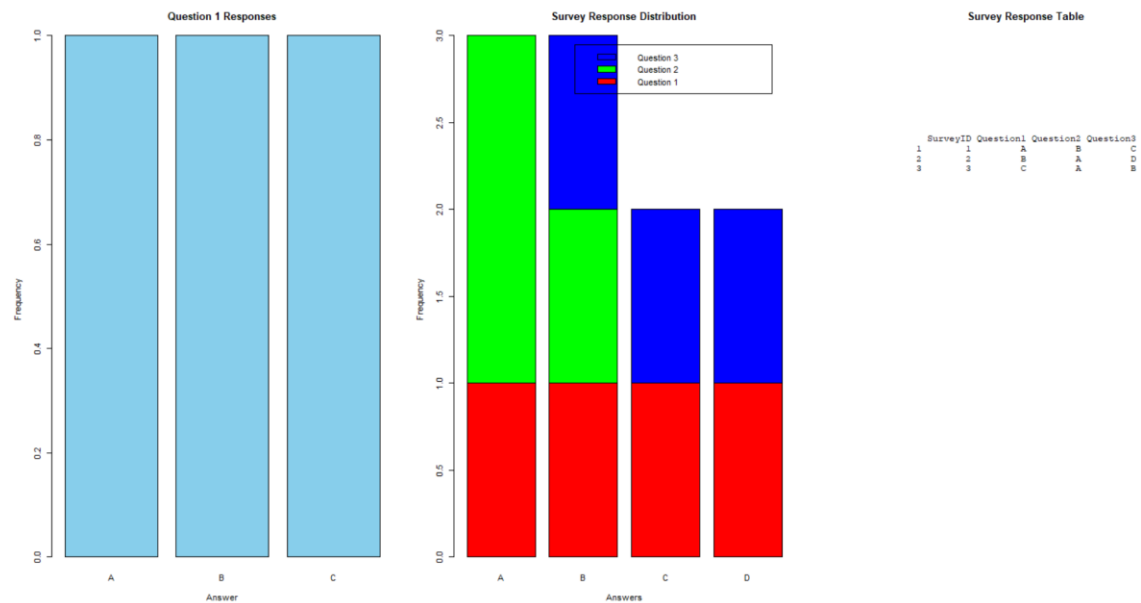
table_data <- data.frame(
  SurveyID=survey_id,
  Question1=question1,
  Question2=question2,
  Question3=question3
)

text(0.5,0.8,
     paste(capture.output(print(table_data)),
           collapse="\n"),
     family="mono")

# Reset Layout
par(mfrow=c(1,1))

```

OUTPUT:



Set 11: Product Category Analysis

R PROGRAM

```

# Product Category Dataset
category <- c("Electronics","Clothing","Appliances")
sales <- c(50000,35000,40000)

# Display 3 outputs together
par(mfrow=c(1,3))

# 1. Funnel Chart
barplot(sort(sales, decreasing=TRUE),

```

```

    horiz=TRUE,
    col="skyblue",
    names.arg=category[order(sales, decreasing=TRUE)],
    main="Sales Conversion Funnel",
    xlab="Sales")

# 2. Table
plot.new()
title("Product Category Table")

table_data <- data.frame(
  Category=category,
  Sales=sales
)

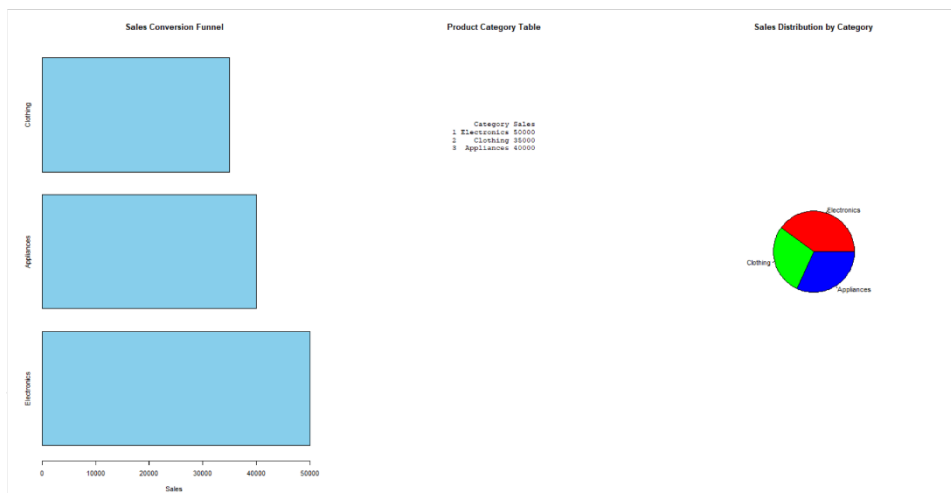
text(0.5,0.8,
     paste(capture.output(print(table_data)),
           collapse="\n"),
     family="mono")

# 4. Pie Chart
pie(sales,
    labels=category,
    col=rainbow(length(category)),
    main="Sales Distribution by Category")

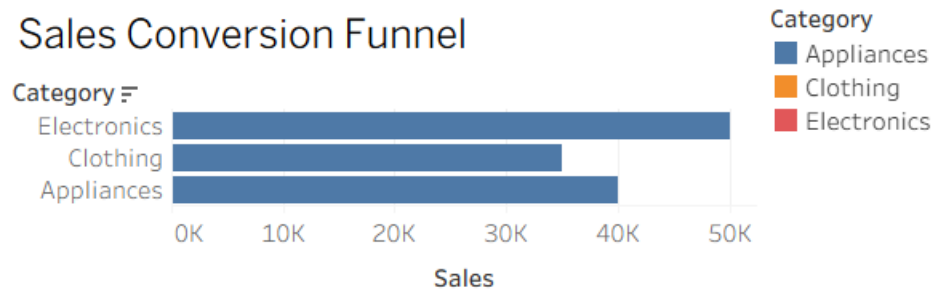
# Reset Layout
par(mfrow=c(1,1))

```

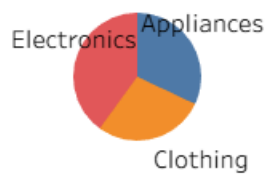
OUTPUT



Sales Conversion Funnel



Sales Distribution by Category



SALES TABLE

Category	
Appliances	40,000
Clothing	35,000
Electronics	50,000

Set 12 : Website Traffic

R PROGRAM

```
# Website Traffic Dataset
date <- c("2023-01-01","2023-01-02","2023-01-03")
page_views <- c(1500,1600,1400)
ctr <- c(2.3,2.7,2.0)
```

```
# Display 3 outputs together
par(mfrow=c(1,3))
```

```
# 1. Line Chart
```



```
plot(page_views,
     type="o",
     col="blue",
     xaxt="n",
     main="Daily Page Views",
     xlab="Date",
     ylab="Page Views")
axis(1, at=1:3, labels=date)
```

2. Bar Chart

```
barplot(ctr,
        names.arg=date,
        col="lightgreen",
        main="Click-through Rate",
        xlab="Date",
        ylab="CTR (%)")
```

3. Table

```
plot.new()
title("Website Traffic Table")
```

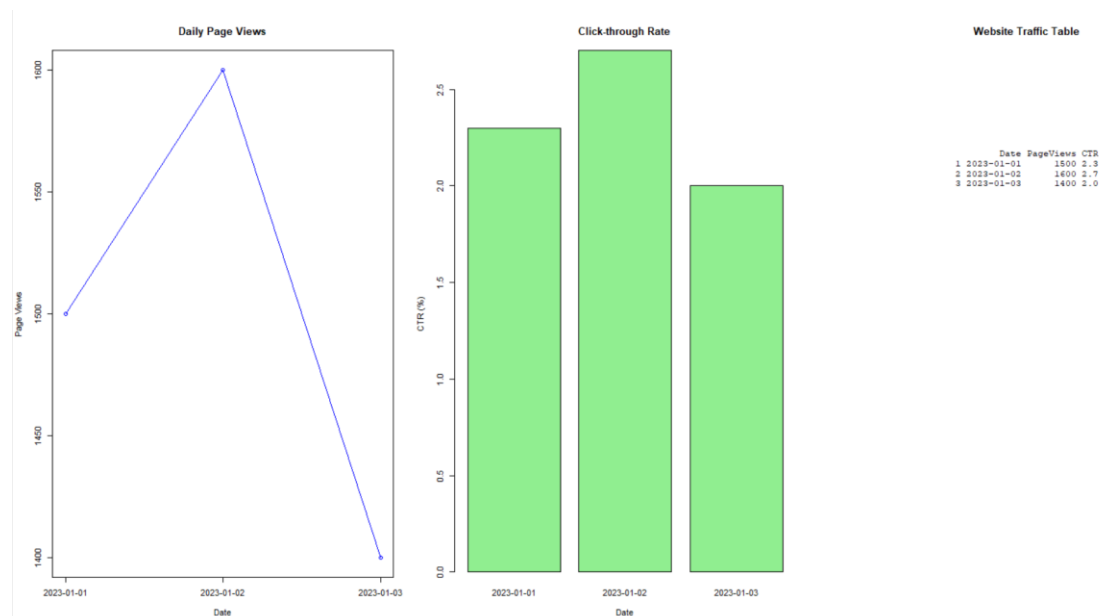
```
table_data <- data.frame(
  Date=date,
  PageViews=page_views,
  CTR=ctr
)
```

```
text(0.5,0.8,
     paste(capture.output(print(table_data)),
           collapse="\n"),
     family="mono")
```

Reset Layout

```
par(mfrow=c(1,1))
```

OUTPUT



Set 13: Geographic Data

R PROGRAM:

```
# Geographic Data Dataset
city <- c("City A","City B","City C")
population <- c(500000,700000,600000)
temperature <- c(75,68,80)
elevation <- c(1000,800,1200)

# Display 3 outputs together
par(mfrow=c(1,3))

# 1. Map Chart (Approximation using Scatter Plot)
plot(population, elevation,
     pch=19,
     col="blue",
     cex=2,
     main="City Distribution",
     xlab="Population",
     ylab="Elevation")

text(population, elevation,
     labels=city,
     pos=3)

# 2. Scatter Plot
plot(temperature, population,
     pch=19,
     col="red",
     main="Temperature vs Population",
     xlab="Average Temperature",
     ylab="Population")

abline(lm(population ~ temperature),
       col="blue",
       lwd=2)

# 3. Table
plot.new()
title("Geographic Data Table")

table_data <- data.frame(
  City=city,
  Population=population,
  AvgTemperature=temperature,
  Elevation=elevation
)

text(0.5,0.8,
     paste(capture.output(print(table_data)),
           collapse="\n"),
     family="mono")

# Reset Layout
par(mfrow=c(1,1))
```

```
# Interpretation
cat("Interpretation:\n")
cat("The scatter plot shows the relationship between average temperature and population.\n")
cat("No strong trend is observed because the dataset contains only three cities.\n")
```

OUTPUT:

